

INFO 521 · Machine Learning Foundations

Course Map · Spring 2026 · 7.5-week accelerated term · School of Information, University of Arizona

Revised 2026-08-04. Reflects the checkpoint-quiz model and the gating schedule agreed 2026-08-03.

Some things to keep in mind

- 3 credit course at 7.5 weeks = 18 to 24 hours per week. The per-week estimates below total ~156 hours, an average of ~20.8.
- Student engagement runs three ways: with content, with the instructor, and with each other.
- Weeks are pacing units. This term has no fixed calendar dates; deadlines within a week are set per offering.

Course competencies

- MS1: Computational Thinking
- MS2: Data Collection, Analysis, and Interpretation
- MS3: Communication of Analytical Results

Course learning outcomes

At the end of this course, students will be able to:

- CLO1. Identify and evaluate machine learning approaches appropriate for regression, classification, clustering, and dimensionality reduction problems.
- CLO2. Implement machine learning algorithms in Python using industry-standard tools and workflows.
- CLO3. Compare and interpret model performance using appropriate evaluation metrics, validation techniques, and statistical reasoning.
- CLO4. Apply machine learning methods to real-world datasets to develop predictive and descriptive models.
- CLO5. Communicate machine learning results, limitations, and recommendations through written, visual, and technical presentations.

Assignment schema

Five graded categories totalling 100%. Nothing carries partial credit: every deliverable is Pass / Not-Yet against published specifications, and the category letters combine by the weights below.

Category	Weight	What it is	Delivered through	How it is graded	Gating
Project	55%	One project, two scopes declared by end of Wk 2. Structured: Parts 1+2 on one dataset, 9 milestones, Part 1 due Wk 5 (midpoint), Part 2 due Wk 7.5. Self-guided capstone: student-designed, midpoint report Wk 5, final Wk 7.5	Classroom 50 repo	Satisfactory / Not-Yet, one revise cycle	Structured: milestones gated by quizzes / peer review. Capstone: three spec bundles + midpoint report, no gates
Homework	25%	Unit 0 (required) + 6 core units (1, 2a, 2b, 3, 4, 5); best 5 core count — one drop built in	Classroom 50 repo, autograded + human review	Ladder: A = any 5 core, B = 4, C = 3, D = 2; U0 required above E	Correctness must pass for a unit to pass
Peer-engagement loops and discussions	10%	6 graded loops (no Week 5)	D2L discussion + course activity tool	Pass / Not-Yet, all four stages required	Time-bound within the week; not revisable
Checkpoints	9%	5 open-book quizzes: 1.1, 1.2, 1.3, 2.1, 2.3	D2L quiz	80% auto-graded bar; written responses Satisfactory / Not-Yet with one revision	Each gates the milestone due the following week
Prerequisite quiz	1%	1, Week 1	D2L quiz	Self-assessment	None

Two changes from the previous schema are worth flagging. Checkpoints were derivation notebooks peer-reviewed by a teammate; they are now open-book D2L quizzes, and the coding verification that sat beside each derivation merged into Homework Units 1, 3, and 4 and into milestones 2.1, 2.3, and 2.4, so nothing is assessed twice. The per-module worked notebooks were removed from the course entirely.

How the checkpoint gate works

Each checkpoint quiz is due the week its module is taught, and gates the project milestone due the following week. The gate therefore resolves before the work it gates comes due. A milestone cannot be marked Satisfactory until its gating quiz is passed.

Quiz	Quiz due	Derivation it covers	What it gates	Milestone due
Checkpoint quiz 1.1	Week 2	Normal equations	Milestone 1.1	Week 3
Checkpoint quiz 1.2	Week 3	MLE for the weights	Milestone 1.2	Week 4
Checkpoint quiz 1.3	Week 4	Conjugate posterior	Milestone 1.3	Week 5
Checkpoint quiz 2.1	Week 5	Newton–Raphson update for the MAP	Milestone 2.1 and Gate A	Week 6

Quiz	Quiz due	Derivation it covers	What it gates	Milestone due
Checkpoint quiz 2.3	Week 6	k-means objective and PCA	Milestone 2.3 and Gate B	Week 7

Gate A (supervised) is satisfied when milestones 2.1 and 2.2 are both Satisfactory and quiz 2.1 is passed. Gate B (unsupervised) when 2.3 and 2.4 are both Satisfactory and quiz 2.3 is passed. Milestones 1.4 and 2.5 are the syntheses, gated by structured peer review rather than by a quiz. All five quizzes are required on both project scopes; they gate milestones only on the structured scope.

Week-by-week map

One module per week. The final half week carries no new material: it exists so the Project 2 synthesis does not land on top of the Week 7 module.

Module / Unit / Week	Outcomes <i>At the end of this module/unit/week, students will be able to...</i>	Learning Materials <i>Lectures, readings, videos, etc.</i>	Activities / Assignments / Assessments <i>Low-stakes assessments, high-stakes assessments, discussions, projects, etc.</i>
Week 1: Foundations of Machine Learning and Data Preparation <i>Estimated student hours: ~18</i>	– Differentiate supervised, unsupervised, and reinforcement learning paradigms. – Prepare datasets for machine learning analysis. – Implement and evaluate simple linear regression models. <i>Supports: CLO1, CLO2, CLO4 Competencies: MS1, MS2</i>	– Welcome and course overview video – Lecture: M1a Setup and Model – Lecture: M1b Normal Equations – Lecture: Appendix, Matrix Calculus – PML1 Ch. 1 §1.1–1.3 and §1.5; MML Ch. 2; MML Ch. 9 §9.1–9.2	– Introductions discussion forum (Discussion 1) – Instructor welcome video and Q&A – Prerequisite quiz (D2L) – Unit 0 + Unit 1 HW – Week 1 Activity: Least Squares explorer (peer-engagement loop) – The project assigned: both scopes described (structured Parts 1+2, or self-guided capstone)
Week 2: Model Evaluation, Bias–Variance, and Regularization <i>Estimated student hours: ~19</i>	– Evaluate model performance using train/validation splits and k-fold cross-validation. – Explain the bias–variance tradeoff and diagnose under- versus over-fitting. – Apply ridge regularization to control model complexity. <i>Supports: CLO1, CLO3 Competencies: MS1, MS2</i>	– Module overview video – Lecture: M2a Evaluation and Cross-Validation – Lecture: M2b Bias–Variance and Ridge – PML1 Ch. 4 §4.7; PML1 §11.3; MML Ch. 5 (review)	– Discussion 2: the bias–variance tradeoff in your own words – Project scope declared: structured or self-guided capstone (with dataset proposal) – Unit 2a + Unit 2b HW – Week 2 Activity: Bias–Variance and Ridge explorer (peer-engagement loop) – Checkpoint quiz 1.1 (D2L): normal equations. Gates milestone 1.1, due Week 3 – Structured scope: milestone 1.1 in progress
Week 3: Probability, Likelihood, and Maximum Likelihood Estimation <i>Estimated student hours: ~21</i>	– Formulate a model's likelihood and estimate parameters by maximum likelihood (MLE). – Show how MLE under Gaussian noise recovers least-squares regression.	– Module overview video – Lecture: M3a Likelihood and MLE – Lecture: M3b MLE Uncertainty – Lecture: Appendix, Covariance of the Estimator	– Discussion 3: what likelihood buys us over least squares – Unit 3 HW (includes the merged MLE == OLS check) – Checkpoint quiz 1.2 (D2L): MLE. Gates milestone 1.2, due Week 4

Module / Unit / Week	Outcomes <i>At the end of this module/unit/week, students will be able to...</i>	Learning Materials <i>Lectures, readings, videos, etc.</i>	Activities / Assignments / Assessments <i>Low-stakes assessments, high-stakes assessments, discussions, projects, etc.</i>
	Quantify parameter uncertainty using Fisher information and the estimator's covariance. <i>Supports: CLO2, CLO3, CLO4 Competencies: MS1, MS2</i>	PML1 Ch. 2, Ch. 3; §4.1–4.2 (incl. §4.2.5); §4.7; MML Ch. 6 (review)	- Milestone 1.1 due (Least Squares) - No interactive tool this module; the written discussion carries the loop
Week 4: Bayesian Inference <i>Estimated student hours: ~21</i>	- Apply Bayes' rule to combine a prior and likelihood into a posterior. - Use conjugacy (Beta–Binomial and the conjugate Gaussian) to derive closed-form posteriors. - Relate ridge regression to MAP estimation under a Gaussian prior. <i>Supports: CLO1, CLO3, CLO4, CLO5 Competencies: MS2, MS3</i>	- Module overview video - Lecture: M4a Bayesian Updating (Beta–Binomial) - Lecture: M4b The Gaussian Posterior - PML1 §4.6; Think Bayes Ch. 2, Ch. 4; PML1 §11.7; Think Bayes Ch. 18 (supplementary)	- Discussion 4: how the posterior contracts with data - Unit 4 HW (includes the merged grid-posterior check) - Week 4 Activity: Bayesian Updating explorer (peer-engagement loop) - Checkpoint quiz 1.3 (D2L): conjugate posterior. Gates milestone 1.3, due Week 5 - Milestone 1.2 due (Maximum Likelihood)
Week 5: Approximate Inference and Classification <i>Estimated student hours: ~24 (heaviest week: project midpoint)</i>	- Fit logistic regression by maximum likelihood using Newton's method / IRLS. - Explain why binarizing the outcome breaks conjugacy and motivates approximate inference. - Approximate a posterior with the Laplace method and interpret a Metropolis sampling trace. <i>Supports: CLO1, CLO2, CLO3 Competencies: MS1, MS2</i>	- Module overview video - Lecture: M5a Logistic Regression and Newton's Method - Lecture: M5b Laplace Approximation and Sampling - PML1 Ch. 10 §10.1–10.2 (review Ch. 9); §4.6.8; Think Bayes Ch. 19; Optional: MML Ch. 7 (continuous optimization)	- No discussion this week (Project 1 closes); the Week 5 tool runs as lecture support - Unit 5 HW - Week 5 Activity: Logistic Regression explorer (tool and mastery check only) - Checkpoint quiz 2.1 (D2L): Newton–Raphson. Gates milestone 2.1, due Week 6, and Gate A - Milestones 1.3 and 1.4 due (structured scope) - Project midpoint: Part 1 due (structured scope) / midpoint report due (capstone scope) - Part 2 begins (structured scope)
Week 6: Support Vector Machines and Clustering <i>Estimated student hours: ~22</i>	- Explain max-margin classification, support vectors, and the role of kernels (concept level). - Implement and run K-means clustering; diagnose local optima and the choice of k. - Connect K-means to Gaussian mixtures and the EM algorithm (hard versus soft assignment). <i>Supports: CLO1, CLO2, CLO4 Competencies: MS1, MS2</i>	- Module overview video - Lecture: M6a Support Vector Machines - Lecture: M6b K-means Clustering - PML1 Ch. 21 §21.3–21.4; MML Ch. 11 - Optional pointer: PML1 Ch. 17 §17.3 (SVMs)	- Discussion 6: when clusters do and do not match labels - Project second half in progress on both scopes - Week 6 Activity: K-means explorer (peer-engagement loop) - Checkpoint quiz 2.3 (D2L): k-means and PCA. Gates milestone 2.3, due Week 7, and Gate B - Milestones 2.1 and 2.2 due (Gate A resolves) (structured scope)

Module / Unit / Week	Outcomes <i>At the end of this module/unit/week, students will be able to...</i>	Learning Materials <i>Lectures, readings, videos, etc.</i>	Activities / Assignments / Assessments <i>Low-stakes assessments, high-stakes assessments, discussions, projects, etc.</i>
Week 7: Dimensionality Reduction and Course Synthesis <i>Estimated student hours: ~21</i>	– Compute principal components via eigen-decomposition of the covariance matrix. – Interpret variance explained and choose the number of components to retain. – Apply PCA to a real dataset and articulate its limitations. <i>Supports: CLO1, CLO4, CLO5 Competencies: MS2, MS3</i>	– Module overview video and course wrap-up – Lecture: M7a Principal Component Analysis – MML Ch. 4 §4.2; MML Ch. 10; PML1 Ch. 20 §20.1 – Where next: UDL Ch. 2–3, 6–7 (optional; continues in INFO 557)	– Discussion 7: the course arc, loss to likelihood to posterior to approximation to discovery – Week 7 Activity: PCA explorer (peer-engagement loop) – Milestones 2.3 and 2.4 due (Gate B resolves) (structured scope)
Week 7.5: Course close (half week) <i>Estimated student hours: ~10 (half week)</i>	– Synthesize the term's methods into one account of a single cohort, naming what each lens saw and what all of them missed. <i>Supports: CLO4, CLO5 Competencies: MS2, MS3</i>	– No new module. Office hours and synthesis support only.	– The project due (either scope): milestone 2.5 + Part 2, or the capstone build and report – No discussion, no homework, no checkpoint

Notes. Module 3 has no interactive tool, so its written discussion carries that week's peer-engagement loop on its own. Week 5 has no graded discussion because Project 1 closes that week; the tool still runs as lecture support. Six graded loops run over the term. Neural networks are the natural continuation of this material and are covered in INFO 557; they are not a module of this course.